

Jack Burnos - Threads lab sep 15 26

Question 1

- I don't think they overlap because one runs to completion before the next is started.

```
[09:57:32] start user thread 1
[09:57:35] end user thread 1
[09:57:35] start user thread 2
[09:57:38] end user thread 2
[09:57:38] start user thread 3
[09:57:41] end user thread 3
[09:57:41] start user thread 4
[09:57:44] end user thread 4
[09:57:44] start user thread 5
[09:57:47] end user thread 5
Total execution time is 15.0146 seconds
```

Question 2

- Kernel threads do overlap because they all begin at the same time and end at the same time

```
[10:00:05] start kernel thread 1
[10:00:05] start kernel thread 2
[10:00:05] start kernel thread 3
[10:00:08] end kernel thread 1
[10:00:08] end kernel thread 2
[10:00:08] end kernel thread 3
Total execution time is 3.0068 seconds
```

Question 3

- Switching user level threading to yielding control instead of blocking the user thread, letting each user thread have a turn by adding "while(threads):" This made the user threads run at the same time and end at the same time like kernel threads

```
[10:03:19] start user thread 1
[10:03:19] start user thread 2
[10:03:19] start user thread 3
[10:03:19] start user thread 4
[10:03:19] start user thread 5
[10:03:22] end user thread 1
[10:03:22] end user thread 2
[10:03:22] end user thread 3
[10:03:22] end user thread 4
[10:03:22] end user thread 5
Total execution time is 3.0023 seconds
```